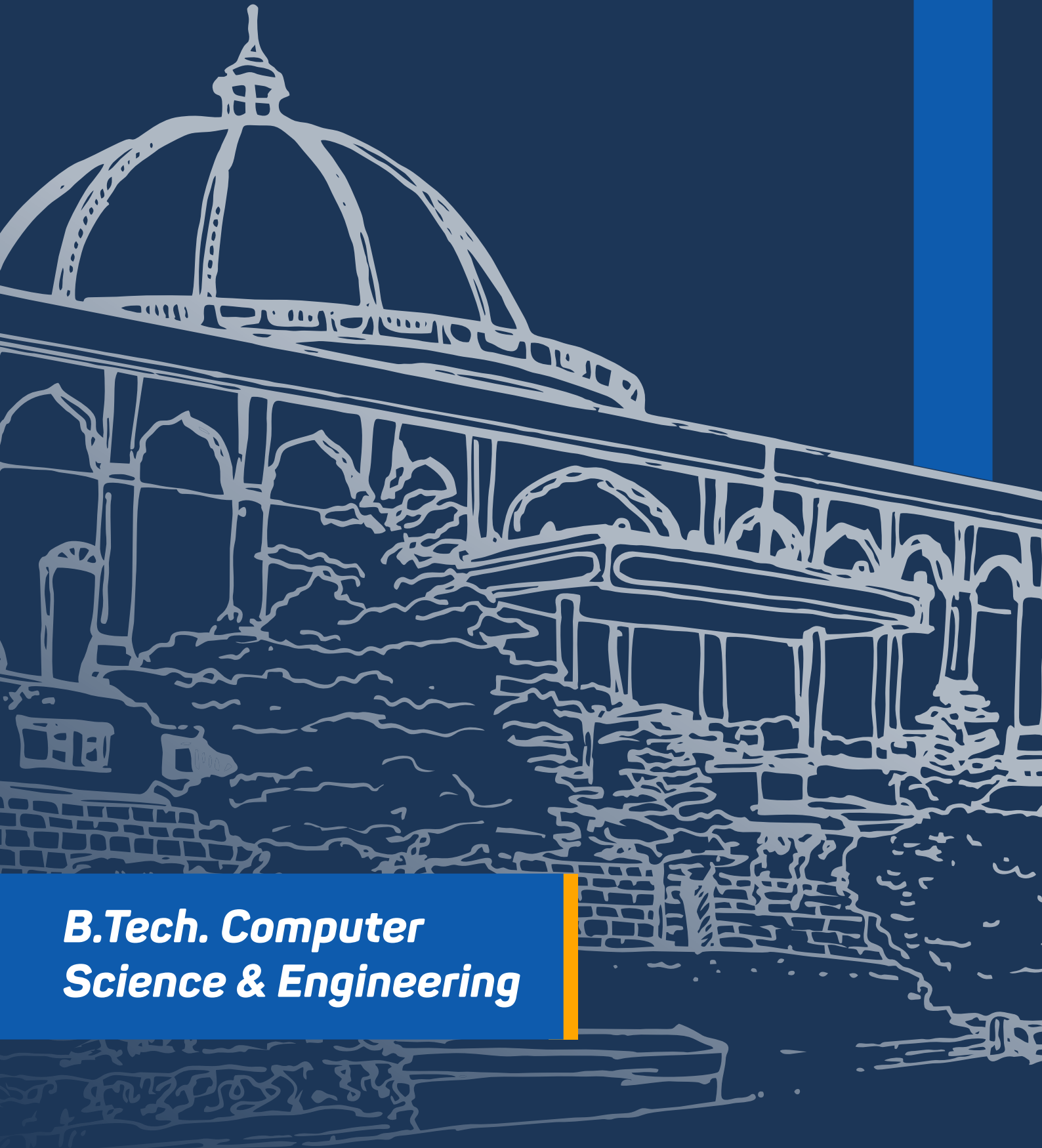




B.Tech. Computer Science & Engineering

Division	Faculty of Engineering and Technology
School Name	School of Computer Science & Engineering
Department Name	Department of Computer Engineering and Technology
Programme Name	B.Tech. Computer Science & Engineering

Semester	Odd (I)	Even (II)	Total Credits
First Year	20	21	41
Second Year	20	21	41
Third Year	22	22	44
Fourth Year	21	14	35

+ + + + + + + + + + + + + + + **COURSE BASKET** + + + + + + + + + + + + + + +

| Course Type | Description |
|----------------------|--|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU |
| Programme Electives | Open electives under the programme allow students to specialise in a particular area connected to their major. |
| University Core | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core. |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| I | PF | Linear Algebra and Differential Calculus | 3 |
| I | PF | Engineering Chemistry | 3 |
| I | PF | Engineering Physics | 3 |
| I | PF | Engineering Graphics | 3 |
| I | PF | Ideas and Innovations in Manufacturing | 1 |
| I | UC | Environment and Sustainability | 1 |
| I | UC | Yoga - I | 1 |
| I | UC | Social Leadership Development Program | 1 |
| I | UC | Python Programming | 2 |
| I | UC | Financial Literacy | 1 |
| I | UC | Indian Constitution | 1 |
| | | Total Credits: | 20 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|---|--|---------------|
| II | PF | Engineering Mechanics | 3 |
| II | PF | Foundations of Programming | 4 |
| II | PF | Discrete Mathematics with Graph Theory | 3 |
| II | Program Capstone Project/ Seminar and Internships | Foundations of Computer Architecture and System Design | 4 |
| II | UC | Yoga - II | 1 |
| II | UC | Co-creation | 1 |
| II | UC | Foundation of Peace | 2 |
| II | UC | Indian Knowledge System (General) | 2 |
| II | UC | Sports | 1 |
| | | Total Credits: | 21 |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| III | PF | Calculus and Numerical Methods | 4 |
| III | PF | Data Structures | 3 |
| III | PF | Data Structures Laboratory | 1 |
| III | PF | Microprocessor, Microcontroller and Applications Lab | 1 |
| III | PM | Microprocessor, Microcontroller and Applications | 3 |
| III | UE | University Elective -I | 3 |
| III | UC | Research Innovation Design Entrepreneurship | 1 |
| III | UC | Spiritual and Cultural heritage: Indian Experience | 2 |
| III | PF | Indian Knowledge System (Sci & Tech) | 2 |
| | | Total | 20 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| IV | University Electives | University Elective -II | 1 |
| IV | Program Capstone Project/
Seminar and Internships | Object Oriented Concepts using C++ | 1 |
| IV | Program Foundation | Probability and Statistics | 3 |
| IV | Program Major | Advanced Data Structures | 4 |
| IV | Program Foundation | Advanced Data Structures Laboratory | 2 |
| IV | Program Major | Database Management Systems | 3 |
| IV | Program Major | Database Management Systems
Laboratory | 2 |
| IV | University Core | Rural Immersion | 3 |
| IV | University Core | Life Transformation Skills | 2 |
| IV | Program Capstone Project/
Seminar and Internships | Seminar | 2 |
| IV | Program Capstone Project/
Seminar and Internships | Cognitive Skills | 2 |
| | | Total Credits: | 21 |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| V | PM | Computer Networks | 3 |
| V | PM | Computer Networks Laboratory | 1 |
| V | PM | Theory Of Computation | 3 |
| V | PM | Internet of Things Laboratory | 1 |
| V | PM | Artificial Intelligence and Expert Systems | 3 |
| V | PM | Artificial Intelligence and Expert Systems Laboratory | 1 |
| V | PM | Software Engineering and Modelling | 3 |
| V | PF | Software Engineering and Modelling Lab | 1 |
| V | UC | Managing Conflicts Peacefully: Tools and Techniques | 2 |
| V | PF | Java Programming | 1 |
| V | UE | University Electives - III | 3 |
| V | PE | Program Elective - I | 4 |
| | | Total Credits: | 22 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| VI | PM | Operating System | 3 |
| VI | PM | Operating System Laboratory | 1 |
| VI | PM | Advanced Algorithms | 3 |
| VI | Program Capstone Project/Seminar and Internships | Data Engineering and Data Visualization | 3 |
| VI | PM | Machine Learning | 3 |
| VI | PM | Machine Learning Laboratory | 1 |
| VI | PM | Full Stack Development Laboratory | 1 |
| VI | Program Capstone Project/Seminar and Internships | Competitive Coding | 1 |
| VI | UC | National Academic Immersion Program | 2 |
| VI | PE | Program Elective - II | 4 |
| | | Total Credits: | 22 |

Semester - VII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|-------------------------------------|---------------|
| VII | PE | Program Elective - III | 4 |
| VII | Program Capstone Project/Seminar and Internships | Capstone Project | 6 |
| VII | PM | Information and Cyber Security | 4 |
| VII | PM | Distributed Computing | 3 |
| VII | PM | System Software and Compiler Design | 4 |
| | | Total Credits: | 21 |

Semester - VIII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|----------------------------|---------------|
| VIII | PE | Program Elective - IV | 4 |
| VIII | Program Capstone Project/Seminar and Internships | Internship | 10 |
| | | Total Credits: | 14 |

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| V | PE - I | Bigdata Technologies | 4 |
| V | PE - I | Computer Graphics and 3D Modelling | 4 |
| V | PE - I | Wireless and Mobile Networks | 4 |
| V | PE - I | Cloud Computing and DevOps | 4 |
| V | PE - I | Digital IC Design | 4 |
| VI | PE - II | Deep Learning | 4 |
| VI | PE - II | Augmented Reality and Virtual Reality | 4 |
| VI | PE - II | Cyber Laws and Cyber Crime | 4 |
| VI | PE - II | Parallel Programming | 4 |
| VI | PE - II | VLSI Design | 4 |
| VII | PE - III | Cognitive Computing and Natural Processing | 4 |
| VII | PE - III | Computer Vision | 4 |
| VII | PE - III | Vulnerability Identification and Penetration Testing | 4 |
| VII | PE - III | User Interface and User Experience Design | 4 |
| VII | PE - III | Blockchain Technology | 4 |
| VIII | PE - IV | Soft Computing | 4 |
| VIII | PE - IV | Unmanned Ariel Vehicle(UAV) and Drone Technology | 4 |
| VIII | PE - IV | VLSI Design Verification and Testing | 4 |
| VIII | PE - IV | 5G and Edge Computing | 4 |
| VIII | PE - IV | Digital Design Automation | 4 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.